State Space Nonlinear Process Modeling:

Identification and Universality

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Abstract

The focus of this research is to create a practical approach for identifying nonlinear models from avai
data. Knowing the true dynamics of a process allows implementation of a better and more efficient control sch
However, this knowledge is not always available, so a valid alternative to obtain a model is to identify the system. For
purpose we use the DABNet (Decoupled A-B Net) structure, which is composed of a decoupled linear dynamic sy
followed by a nonlinear static map. The linear dynamic system is initially spanned by a set of discrete Laguerre systems
then cascaded with a single hidden layer Perceptron. A model reduction technique (linear balancing) is performed o
hidden nodes of the neural network as part of the identification process. In that way, it is possible not only to identif
main time constants, but also to reduce the dimensionality of the Perceptron input. We prove that the DABNet structu
able to approximate every nonlinear, causal, discrete time invariant, multiple-input single-output system with fading mem
The final model consists of a linear state space system whose states, decoupled by input, are mapped by a neural netw
Results concerning the application of the methodology to the approximation of a CSTR and a polymer process are presen

1. Introduction

This paper presents a practical approach for identifying nonlinear dynamic models from process data. There are
main approaches to obtaining a dynamic process model. The first one is the first principle modeling approach, whi
usually costly and highly dependent on the availability of process knowledge. The alternative is the empirical mod
approach, in which the model is identified from the data. While first principle models are appropriate for many purp
(simulation, operator-training, etc.), the identification approach is the most feasible and practical for control purpose
most industrial processes. However, in the identification of a general nonlinear model it is essential to deal with the issu
ensuring *long-term prediction capabilities*, *a feasible and practical identification procedure and, model suitability f*
certain kinds of processes. We address these problems in order to make nonlinear empirical model more reliable.

The first important issue to be considered is ensuring the long-term prediction capabilities of the model. Th
evident in implementations of Model Predictive Control (MPC) techniques, one of the most successful process co
approaches (Morari and Lee, 1997). In MPC, a dynamic model is used to predict the behavior of the system over a f
time horizon in response to a sequence of input changes. The controller is in fact a minimization problem with constr
(Oliveira and Biegler, 1996) which plans a control path for which the prediction path is optimal respect to a set criterion
as set point mismatches and suppression moves. Stability is achieved by choosing the horizon large enough, i.e., larger
the maximum time required to steer the system to the origin from any state (Michalska and Mayne, 1993). In many sys
this horizon may be large, *so models should be able to predict over large horizons without degrading the predi*
performance.

A second very important point is getting a feasible and practical identification procedure. Most industrial proc
that apply MPC employ linear step-response models or finite impulse response models (FIR). Although these approache
widely and successfully used in industry, they require extensive plant testing, which often results in over-parameterizatio
the models (Qin and Badgwell, 1996). For example the dynamics of a first order single input/single output (SISO) pro
which can be represented with only three parameters (gain time constant, and dead-time) in a parametric form, typi
requires 30 to 120 coefficients. Alternative MPC implementations using Neural Networks models are describe
Hernández and Arkun (1990), Temeng *et al* (1992) and Ishida and Zhan (1995). A wavelet-based model is used in H
Juarez and Kantor (1992), while the use of Recurrent Neural Networks may be found in Williams and Zipser (1
Narendra and Parthasarathy (1991) and Chen and Weingand (1994). Recurrent Neural Networks provide a reason
solution to long-term prediction problems, but training such models is very difficult. *So, efficient and feasible tra*
methods are needed.

A third important issue is to prove the approximation capabilities of the models. Great efforts have been made i development of nonlinear system approximation methods using conventional means (De Figueiredo and Dwyer, 1 Leontaritis and Billings, 1985; Korenberg and Paarmann, 1991; Pottmann *et al*, 1993; Boutayeb and Darouach, 1995) ar novel approaches like Neural Networks (Moody and Darken, 1988; Lapedes and Farber, 1989; Narendra and Parthasar 1991; Lin Lin *et al*, 1995). Among all the methods, there are only a few available results proving that a spe identification model may approximate a certain kind of nonlinear system. *So, determining the theoretical capabilities feasible model and to which systems they apply, is very important.*

In this paper, we present a methodology to obtain models for a class of nonlinear systems in terms of a l dynamic part followed by a nonlinear static map using input-output data. The linear dynamic system is initially spanned set of discrete Laguerre systems, and then cascaded with a single hidden layer Perceptron as can be seen in Figure **1-a** previous work (Sentoni *et al*, 1996), it was shown that a nonlinear combination of discrete Laguerre systems is ab approximate every single-input nonlinear discrete system having fading memory. The *fading memory* concept (Boyd Chua, 1985) is an important part of the characterization and is still the subject of current research (Shamma and Zhao, 1 Matthews and Moschytz, 1994; Partington and Mäkilä, 1996, Sentoni 1997). After the initial span of the linear layer Laguerre systems, a model reduction technique (a linear balancing, Zhou *et al*, 1996) is performed on the hidden nodes o neural network as part of the identification process. The balancing is performed in such a way that the linear state s representation is de-coupled by blocks. Doing so, it is possible not only to identify the main time constants, but al reduce the dimensionality of the Perceptron input space. In addition, the non-controllable/non-observable modes removed from the model. The final *DABNet* model (De-coupled A-B matrices Neural Network) consists of a sparse l state space system whose states, de-coupled by blocks, are mapped by a neural network. This de-coupled represent provides insight into the final model, especially since, in a practical identification scheme, the individual state space mat can themselves be represented as loosely coupled first and second order sections. The *DABNet* structure is well suited embedded into a MPC Strategy, providing better long-term predictions than a NARX model in which the output error fed back into model input and amplified (Tsoi and Back, 1994). Besides, *DABNet* models show all of the following feat long-term prediction capabilities with minimal parameterization, practical identification that derive information for pro control and ability to identify a large class of real world processes.

This paper is structured as follows. In the Nonlinear Model Representation section we present the main re concerning the model structure while the universality proof is left for the appendix. The Identification and Model Redu section presents the bulk of the identification process and a summary of the methodology. In the Application Exam section, we apply the proposed methodology to the approximation of typical process unitis: a CSTR and a contin

polymerization process. Finally, we present our conclusions and the future work arising from this study.

2. Nonlinear Model Representation

A process can be represented by an operator, that is a function, which maps input signals into output signals. I are interested in discrete systems, those signals become discrete sequences. Let $\boldsymbol{Z}$ denotes the integers and $\boldsymbol{R}^p$ the spa real p-vectors, $p > 0$, with the norm $|\mathbf{x}| = \max_{1 \le j \le p} |x_j|$. In this context, a sequence is an element of $\boldsymbol{l}_p^\infty(\boldsymbol{Z})$, the spac bounded p-valued vector sequences defined on $\boldsymbol{Z}$, with the norm $\|\mathbf{u}\|_\infty := \sup_{k \in \boldsymbol{Z}} |\mathbf{u}(k)|$. This space becomes $\boldsymbol{l}_p^\infty(\boldsymbol{Z}_+$ sequences defined only for the non-negative integers, $\boldsymbol{Z}_+$, or $\boldsymbol{l}_p^\infty(\boldsymbol{Z}_-)$ for sequences defined for the non-positive integers To clarify the notation, matrices are denoted in bold uppercase while functions are denoted in italic lowercase if they scalars or in bold its vector valued. Thus, **u** represents a vector or a vector sequence in accordance with the context, $\mathbf{u}(k)$ represents a value of the sequence at the particular time k. Moreover, an *operator* $N : \boldsymbol{l}_p^\infty(\boldsymbol{Z}) \to \boldsymbol{l}_p^\infty(\boldsymbol{Z})$ is a fun mapping sequences into sequences while, a *functional*, $F : \boldsymbol{l}_p^\infty(\boldsymbol{Z}_-) \to \boldsymbol{R}$, is a function mapping sequences into a nur When we talk about identifying a process, we are implicitly talking about approximating its associated operator.

We are interested in approximating the class of nonlinear, causal, discrete time-invariant, multiple-input si output fading memory systems using the structure shown in Figure **1-b**. Intuitively, a system has fading memory Appendix for a formal definition) if two signals that are close in the recent past (but not necessarily close in the remote yield present outputs that are close. For dynamical systems, the fading memory concept is related to the uniqueness of st state. For this kind of system, we propose the structure showed in Figure **1-b** which is initially spanned using Lag systems as depicted in Figure **1-a**. It is well known that a great variety of linear systems can be approximated by a l combination of Laguerre systems (Gu *et al*, 1989). The next theorem follows and extends this idea.

Theorem: DABNet Structure

Let $\boldsymbol{K} := \{\mathbf{u} \in \boldsymbol{l}_p^\infty(\boldsymbol{Z}) | 0 < \|\mathbf{u}\| \le c_1 < \infty\}$ be any ball in $\boldsymbol{l}_p^\infty(\boldsymbol{Z})$, and suppose N is any time-invariant ope $N : \boldsymbol{l}_p^\infty(\mathbf{Z}) \to \boldsymbol{l}^\infty(\boldsymbol{Z})$ with fading memory on $\boldsymbol{K}$. Then, given any $\varepsilon > 0$ there are a set of Laguerre oper $\{L_1^1(\cdot), \cdots, L_{m_1}^1(\cdot), \cdots, L_1^p(\cdot), \cdots, L_{m_p}^p(\cdot)\}$ and a Single Hidden Layer Perceptron $NN(\cdot)$ characterized by $(\mathbf{w}2, v2, \mathbf{W}1, \mathbf{v}1)$ such for all $\mathbf{u} \in \boldsymbol{K}$

$$\left\| N\mathbf{u} - \hat{N}\mathbf{u} \right\| < \varepsilon,$$

where the *DABNet* model $\hat{N}$ is given by $\hat{N}\mathbf{u}(k) = NN\left(L_1^1\mathbf{u}(k), \cdots, L_{m_1}^1\mathbf{u}(k), \cdots, L_1^p\mathbf{u}(k), \cdots, L_{m_p}^p\mathbf{u}(k)\right)$.

Proof: For a proof of this theorem, see Appendix A.

This theorem ensures that a certain class of non-linear systems can be approximated by a "nonlinear combinatio Laguerre systems to any degree of accuracy $\varepsilon > 0$. This approximation holds over a bounded input signal domain ***K***, in agreement with any physical system. The nonlinear combination is performed by a Single Hidden Layer Perceptron N characterized by the parameters $(\mathbf{w}2, v2, \mathbf{W}1, \mathbf{v}1)$. The use of this neural network allows the utilization of order redu techniques for pruning the dimension on the linear part, initially spanned by the Laguerre systems.

Laguerre Systems

The advantage of the Laguerre models is that they provide a stable parameterization for a linear system by u orthonormal basis functions generated with the same pole. For a system with *p* inputs a set of Laguerre systems can de in the complex z-plane as:

$$L_i^j(\cdot) = \frac{\sqrt{\eta_j}\, z^{-d_j+1}}{z - a_j}\left[\frac{1 - a_j z}{z - a_j}\right]^i,\ i: 0,\ldots,\infty,\ j: 1:,..,p,$$

where

$L_i^j(z)$ = *Z*-transform of $l_i^j(k)$, the *i*-th order discrete Laguerre system for the *j*-th input component

a_j = generating pole associated to the *j*-th input, such that $|a_j| < 1$

$\eta_j = 1 - a_j^2$, just a constant

$d_j = 1$, unless there is a pure time delay for the *j*-th input for which $d_j > 1$

It is well known that a linear combination of Laguerre systems (2) can approximate a great variety of linear sys (Heuberger *et al*, 1993; Wahlberg, 1991; Gu *et al*, 1989). The main feature in those schemes is that the linear system ca expressed as a linear combination of orthonormal systems generated with the same pole. The only necessary prev knowledge of the systems is reflected in choosing this pole a_j, closely related to its main time constant τ_j (Zervos 1988):

$$a_j = 1 - T/\tau_j$$

The dominant time constant τ_j is relating the output with the *j*-th input while *T* is the sampling time. In this approach identification structure may be easily increased by augmenting the number of Laguerre systems. Besides, the identific task is robust respect to the choice of sampling time *T*, as well as to the choice of model order (Wahlberg, 1 Furthermore, the identification process is not very sensitive with respect to this parameter (Wahlberg, 1994). Notice, tha

set (2) has an infinite number of terms, and for a practical application this number should be finite. To do this we take Laguerre systems relating the output to the *j*-th input generated with the pole a_j, for $j:1:,..,p$. Next, we need to express set in a state space form for then applying a model order reduction technique. Here we represent a linear dynamic system general way, by the quadruple of matrices $[\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}]$ defining its dynamics. Therefore, the quadruple $[\mathbf{A}_j,\mathbf{b}_j,\mathbf{C}$ denotes the m_j systems associated with the *j*-th input expressed in state space form. Those matrices can be expressed a composition of a low-pass section, $\sqrt{\eta_j}(z-a_j)^{-1}$, in series with (m_j-1) all-pass sections $(1-za_j)/(z-a_j)$. Pos realizations are $\left[a_j,\sqrt[4]{a_j^2\eta_j},\sqrt[4]{a_j^2\eta_j},0\right]$for the low-pass and $\left[a_j,\sqrt{\eta_j},\sqrt{\eta_j},-a_j\right]$ for the all-pass systems respectively. Then state space representation $[\mathbf{A}_j,\mathbf{b}_j,\mathbf{C}_j,\mathbf{0}]$ is obtained by performing simple systems operations (see for example pp. Zhou et al, 1996). The outputs of the whole set of the Laguerre systems $\hat{\mathbf{y}}(k)$ are the outputs of the following sy expressed in state space from:

$$\begin{aligned}\hat{\mathbf{x}}(k+1)&=\hat{\mathbf{A}}\hat{\mathbf{x}}(k)+\hat{\mathbf{B}}\mathbf{u}(k)\\ \hat{\mathbf{y}}(k)&=\hat{\mathbf{C}}\hat{\mathbf{x}}(k)\end{aligned},$$

where $m=\sum_{j=1}^{p}m_j$, $\hat{\mathbf{A}}:m\times m$, $\hat{\mathbf{B}}:m\times p$ and $\hat{\mathbf{C}}:m\times m$. Notice the particular structure of the $\hat{\mathbf{A}}$ and $\hat{\mathbf{B}}$ matrices:

$$\hat{\mathbf{A}}=\begin{bmatrix}\mathbf{A}_1 & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \ddots & \cdots & \mathbf{0} & \mathbf{0}\\ \vdots & \vdots & \mathbf{A}_j & \vdots & \vdots\\ \mathbf{0} & \mathbf{0} & \cdots & \ddots & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{A}_p\end{bmatrix},\ \hat{\mathbf{B}}=\begin{bmatrix}\mathbf{b}_1 & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \ddots & \cdots & \mathbf{0} & \mathbf{0}\\ \vdots & \vdots & \mathbf{b}_j & \vdots & \vdots\\ \mathbf{0} & \mathbf{0} & \cdots & \ddots & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{b}_p\end{bmatrix}$$

showing a balanced realization with independent states $\hat{\mathbf{x}}_j(k):1\times m_j$ (de-coupled by blocks). In a linear system, these s are linearly combined to obtain a linear model. In the *DABNet* structure, these states are nonlinearly mapped (preserving nice structure) by a single hidden layer Perceptron.

A single hidden layer Perceptron $NN(\cdot)$ characterized by $(\mathbf{w}2,v2,\mathbf{W}1,\mathbf{v}1)$, perhaps the most used neural netw completes the *DABNet* structure by mapping the states of *Eq.* **4** into the output $\tilde{y}(k)=NN[\hat{\mathbf{x}}(k)]$:

$$\tilde{y}(k)=\mathbf{w}2\Phi\left[\mathbf{W}1\,\hat{\mathbf{C}}\,\hat{\mathbf{x}}(k)+\mathbf{v}1\right]+v2$$

$\mathbf{W}1:[n1\times m]$ is the input weight matrix with $n1$ neurons; $\mathbf{v}1:[n1\times 1]$ is the input bias vector; $\mathbf{w}2:[1\times n1]$ is the output w vector, and $v2$ is the output bias. The activation function $\Phi(s_1,s_2,...,s_{n1})=(\phi(s_1),\phi(s_2),...,\phi(s_{n1}))$ is any permitted sigm activation function $\phi(\cdot)$ applied component-wise. This completes the description of the proposed structure.

3. Identification and Model Reduction

In this section, we reduce the dimension of the linear dynamic part of the *DABNet* structure by keeping the u de-coupled matrix structure. In a first step, the model is adjusted using Laguerre systems because little knowledge is req from a dynamic point of view. This is very important to avoiding a complex nonlinear fitting algorithm, like traditional for these kind of structures (Boutayeb and Darouach, 1995). However, this is at a small added cost: the dimension o state vector $\tilde{\mathbf{x}}(k)$ is bigger than if were using all the (unknown) dynamic information. What can be done to reduc dimension of this vector? Let us rewrite the set of equations for the *DABNet* model:

$$\hat{\mathbf{x}}(k+1) = \hat{\mathbf{A}}\hat{\mathbf{x}}(k) + \hat{\mathbf{B}}\mathbf{u}(k)$$
$$\tilde{\mathbf{y}}(k) = \mathbf{w}2\Phi\left(\hat{\mathbf{W}}1\,\hat{\mathbf{x}}(k) + \mathbf{v}1\right) + v2$$

where $\hat{\mathbf{W}}1 = \mathbf{W}1\,\hat{\mathbf{C}} = \left[\hat{\mathbf{W}}1_1 \ldots \hat{\mathbf{W}}1_j \ldots \hat{\mathbf{W}}1_p\right]$, $\hat{\mathbf{W}}1_j : n1 \times m_j$ for $n1$ hidden neurons and m_j Laguerre systems. Notice tha system $\left[\hat{\mathbf{A}}_j, \hat{\mathbf{b}}_j, \hat{\mathbf{W}}1_j, \mathbf{0}\right]$ forms a linear system with one output for every hidden neuron, over which the sigmoidal activ function is applied. Because this system is linear, all the theory of linear model reduction can be applied. In that way final model is obtained after balancing and truncating the linear system $\left[\hat{\mathbf{A}}_j, \hat{\mathbf{b}}_j, \hat{\mathbf{W}}1_j, \mathbf{0}\right]$ to reduce the dimension o system. The objective of this step is to reduce the dimension of the vector $\hat{\mathbf{x}}_j(k)$, retaining the highest possible quanti information in the reduced system. One way to do this is to generate a balanced realization of the linear sy $\left[\hat{\mathbf{A}}_j, \hat{\mathbf{b}}_j, \hat{\mathbf{W}}1_j, \mathbf{0}\right]$ (Zhou *et al*, 1996) and then truncate it.

The observability Grammian $\mathbf{P}_j$ and the controllability Grammian $\mathbf{Q}_j$ (for $j : 1,..., p$) are matrices that solv following discrete-time Lyapunov equations:

$$\mathbf{P}_j - \left(\hat{\mathbf{A}}_j\right)^T \mathbf{P}_j\, \hat{\mathbf{A}}_j = \left(\hat{\mathbf{W}}1_j\right)^T \hat{\mathbf{W}}1_j$$
$$\mathbf{Q}_j - \hat{\mathbf{A}}_j\, \mathbf{Q}_j \left(\hat{\mathbf{A}}_j\right)^T = \hat{\mathbf{b}}_j \left(\hat{\mathbf{b}}_j\right)^T$$

uniquely for the linear system $\left[\hat{\mathbf{A}}_j, \hat{\mathbf{b}}_j, \hat{\mathbf{W}}1_j, \mathbf{0}\right]$ (Zhou *et al*, 1996). Additionally, there exists a similarity transformatio (Zhou *et al*, 1996) in which the system is in balanced form, that is:

$$\mathbf{P}_{j_{bal}} = \mathbf{Q}_{j_{bal}} = \Sigma_j = diag\left(\sigma_{j1},...,\sigma_{jm_j}\right);$$

where $\sigma_1 \geq ... \geq \sigma_m$ are the Hankel singular values. Now, a balanced realization can be computed simply as

$$\mathbf{A}_{j_{bal}} = \mathbf{T}_j \hat{\mathbf{A}}_j \left(\mathbf{T}_j\right)^T,\ \hat{\mathbf{b}}_{j_{bal}} = \mathbf{T}_j \hat{\mathbf{b}}_j,\ \hat{\mathbf{W}}1_{j_{bal}} = \hat{\mathbf{W}}1_j \left(\mathbf{T}_j\right)^T.$$

Small Hankel values ($\sigma_{jr} >> \sigma_{jr+1}$) indicate that the system can be reduced to an *r*- order model by direct truncation as:

$$\overline{\mathbf{A}}_j = [\mathbf{I}_r\, 0]\mathbf{A}_{j_{bal}},\ \overline{\mathbf{b}}_j = [\mathbf{I}_r\, 0]\overline{\mathbf{b}}_{j_{bal}},\ \overline{\mathbf{W}}1_j = \mathbf{W}1_{j_{bal}}\begin{bmatrix}\mathbf{I}_r\\ 0\end{bmatrix}$$

The final reduced system:

$$\begin{aligned}\overline{\mathbf{x}}(k+1) &= \overline{\mathbf{A}}\,\overline{\mathbf{x}}(k) + \overline{\mathbf{B}}\,\mathbf{u}(k)\\ \overline{\mathbf{y}}(k) &= \overline{\mathbf{w}}2\ \Phi\left(\overline{\mathbf{W}}1\ \overline{\mathbf{x}}(k) + \overline{\mathbf{v}}1\right) + \overline{v}2\end{aligned}.$$

where $\overline{\mathbf{W}}1 = \left[\overline{\mathbf{W}}1_1 \ldots \overline{\mathbf{W}}1_j \ldots \overline{\mathbf{W}}1_p\right]$. The matrices $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$, have the same structure than $\hat{\mathbf{A}}$ and $\hat{\mathbf{B}}$ in *Eq.* **5,** are bui

using $\overline{\mathbf{b}}_j$ $\overline{\mathbf{A}}_j$ from *Eq.* **12**.

Proposed Strategy: Summary

1. *Expand the System into a Laguerre Basis*
   - Select m_j Laguerre systems (Eq. **2**) for each input, typically $5 \le m_j \le 10$ are enough.
   - Select each pole of the Laguerre basis to be determined as a_j (*Eq.* **3**).
   - Calculate the state space representation $\left[\hat{\mathbf{A}}, \hat{\mathbf{B}}, \hat{\mathbf{C}}, \mathbf{0}\right]$ for the whole set of Laguerre systems (*Eq.* **4**).
2. *Train the Neural Network*
   - Given *N* samples $(\mathbf{u}(k), y(k))$ drawn from the system, evaluate the Laguerre states $\tilde{\mathbf{x}}(k)$ (*Eq.* 7**.1**).
   - With $\tilde{\mathbf{x}}(k)$ as input and the output $y(k)$, fit the model to get the parameters $\left(\mathbf{w}2, \mathbf{b}2, \tilde{\mathbf{W}}1, \mathbf{b}1\right)$ (*Eq.***7.2**).
3. *Balance the Model over the Hidden Nodes*
   - Find $\mathbf{P}_j$ and $\mathbf{Q}_j$ Grammians by solving the Lyapunov equations (**8**) and (**9**).
   - Find $\mathbf{T}_j$ that makes the system balanced $\mathbf{P}_{j_{bal}} = \mathbf{Q}_{j_{bal}}$, and the singular values $\left(\sigma_{j_1}, \ldots, \sigma_{j_{m_j}}\right)$ (*Eq.***10**).
   - Find $\mathbf{A}_{j_{bal}}$, $\hat{\mathbf{b}}_{j_{bal}}$, $\tilde{\mathbf{W}}1_{j_{bal}}$ that balance the system (*Eq.* 11).
4. *Truncate the model*
   - If $\sigma_{j_r} >> \sigma_{j_{r+1}}$ (typically $\sigma_{j_r} > 10\,\sigma_{j_{r+1}}$), reduced to an *r*- order model by direct truncation using **12**.
5. *With the new dynamics* $\left(\overline{\mathbf{A}}, \overline{\mathbf{B}}\right)$*, retrain the neural network for the best fit to get the model*

$$\begin{aligned}\overline{\mathbf{x}}(k+1) &= \overline{\mathbf{A}}\,\overline{\mathbf{x}}(k) + \overline{\mathbf{B}}\,\mathbf{u}(k)\\ \overline{\mathbf{y}}(k) &= \overline{\mathbf{w}}2\ \Phi\left(\overline{\mathbf{W}}1\ \overline{\mathbf{x}}(k) + \overline{\mathbf{v}}1\right) + \overline{v}2\end{aligned}$$

Practical Considerations about Modeling and Balancing

As a practical methodology, the Laguerre basis can be optionally replaced by a set of first/second order sections

cover the practical frequency ranges for each input. In addition, each reduced A-B section can itself be decomposed standard first and second order sections. This can be done both from a practical point of view and also, approximately, a theoretical point of view. In practice it is typically unnecessary to consider multiple poles greater than order two. therefore, a Jordan Canonical Form comprising of first and second sections can be used for each input section. Moreov more reduced model can be obtained even if the balancing and reduction procedure is applied to $\left[\hat{\mathbf{A}}, \hat{\mathbf{B}}, \hat{\mathbf{W}}1, \mathbf{0}\right]$, th considering the global dynamics. However, in this way the structure in the matrices $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$ is lost: these matrices bec full matrices.

4. Application Examples

In this section, we illustrate the feasibility of the identification methodology applied to typical process units. S experimental conditions apply to all of the cases and will be stated here. The input sequence uses a probability of switc factor P_s (Hernández, 1993; see also Mane and Doyle, 1997; and Barker, 1993), that corresponds to the probability tha input changes at the end of any sampling interval. If the input is changed, its new values are drawn from a uni distribution with a probability P_s. This probability P_s is adjusted to provide a signal that excites the system evenly ove frequency range of interest. The output signals are integrated using a Gear method taking into account a zero order ho the input. In all the experiments the input/output signals have been scaled between [-1,1] previous to the identification re-scaled after the identification process has taken place. Every particular condition is stated within the proper example the first example, we identify a continuous stirred tank reactor.

Continuous Stirred Tank Reactor

The following set of equations is a first principle model of a two-tank pH neutralization process depicted in F **2**. This is a highly non-linear process that has been widely used as a benchmark problem for non-linear process mod (Zhao *et al*, 1997). The equations defining the mass balance, the acetate balance and the sodium balance in tank 1 an mass balance, the acetate balance and the sodium balance in tank 2 are:

$$A_1\, dh_1/dt = Q_1 + Q_2 + Q_3 - c_{V1}\sqrt{h_1} \quad (1)$$

$$V\, d\alpha_1/dt = Q_1 C_A + Q_2 C_A - c_{V1}\alpha_1\sqrt{h_1} \quad (2)$$

$$V\, d\beta_1/dt = Q_3 C_B - c_{V1}\beta_1\sqrt{h_1} \quad (3)$$

$$A_2\, dh_2/dt = Q_5 + Q_6 + Q_7 + c_{V1}\sqrt{h_1} - c_{V2}\sqrt{h_2} \quad (4)$$

$$V\, d\alpha_2/dt = Q_5 C_A + Q_6 C_A + c_{V1}\alpha_1\sqrt{h_1} - c_{V2}\alpha_2\sqrt{h_2} \quad (5)$$

$$V\, d\beta_2/dt = Q_7 C_B + c_{V1}\beta_1\sqrt{h_1} - c_{V2}\beta_2\sqrt{h_2} \quad (6)$$

where the state variables are:

- $\alpha_{1,2}$: acetate concentration coefficients for tanks 1,2
- $\beta_{1,2}$: sodium concentration coefficients for tanks 1,2
- $h_{1,2}$: height in tanks 1,2

and Qi are the volumetric flowrates shown in Figure 2, C_A and C_B are the acetate and sodium inlet concentrat respectively, and c_{v1} and c_{v2} are the valve constants for tanks 1 and 2, respectively. The whole system is then compl defined with the addition of the acetic acid and water equilibrium equations and the electroneutrality equa $[AC^-]_i[H^+]_i = K_A[HAC]_i$, $[H^-]_i[OH^-]_i = K_W$, $\beta_i + [H^+]_i = [OH^+]_i + [AC^-]_i$ for both tanks (i=1, 2) and $\alpha_i = [HAC] + [AC^-]$ and $\beta_i = N_a^+$ values. The values for the constants and parameters for this model can be found i Table 1. Finally, the definition for the pH is

$$pH = \log_{10}\left[H^+\right]$$

Experimental Conditions. The aim of this example is to identify a non-linear dynamic model of the pH value in th tank as a function of acid and base flow-rates in the 1st tank while maintaining constant the values of the variables in th tank. This is a very nonlinear problem as can be appreciated from the Figure **3** that illustrates the steady state of the pH versus the input stream Q_1. As usual two data sets were generated to identify the system: a training and test set us probability P_s=0.04. The outputs were generated by integrating those inputs using the set of equations **(20)** with a sam time of $T = 0.1$ sec. The former set was used to fit the model, while the latter one was used to validate the model. DABNet model is built to reproduce pH_2, the pH value in the 2nd tank, using Q_1, Q_2 and Q_3 as input variables and kee constant the variables Q_5, Q_6 and Q_7. As a way to show the relative robustness of this approach against choosing bad v for the Laguerre poles, we set them to $a_1 = a_2 = a_3 = 0.5$ (just in the middle of the valid range), while the dead-times selected to be: $d_1 = 2, d_2 = 2, d_3 = 1$. Six Laguerre basis terms were used for each input for the initial expansion of the l dynamic part. For the neural network, we arbitrarily chose 10 hidden nodes without attempting to optimize these paramc After the application of the methodology the system was reduced to a second order system per input. The Figure **4** show approximation results for the DABNet model and for a linear model. The linear model was adjusted using a PLS (P Least Squares) strategy, being composed by a six state linear state space model. Figure **4-a** shows the DA approximation and the real output of the system for the training and test sets, while Figure **5-a** shows the training and errors for the final DABNet model. For comparison purposes, Figure **4-b** shows the linear model approximation and the output of the system, while Figure **5-b** shows the training and test errors for the linear model. Note that the nonlinear m performs much better than the linear one.

It is interesting to address that the steady state curve of the DABNet fairly fits closely the steady state curve o system even though it its very nonlinear. The small discrepancy in the low pH range (pH<6) is due to the lack of inform used to train the model in this range. As can be appreciated from the Figure **4**, the data used to fit the model was for range greater than 6. Here the DABNet model is extrapolating in this zone.

Continuous Polymerization Reactor Process

In this example, we will apply the proposed methodology to identify the conversion factor of a contin polymerization reactor shown in the Figure **6**. The process consists of a polymerization solution of methyl-methacr (MMA), initiated by azo-bis-isobutotyronitrile (AIBN), in which toluene is used as the solvent (Soroush and Kravaris, 1 This model is based on the standard free-radical reaction kinetic mechanism and under standard assumptions, the rea rates for the monomer and initiator are given by:

$$R_m = -C_m \xi_0 \left(k_p + k_{f_m}\right)$$

where $\xi_0 = \left(2f\, k_i C_i / k_t\right)^{0.5}$. The overall propagation rate constants for the gel and glass effects are:

$$k_t = \begin{cases} k_{t_0}\alpha_{t_{11}} \exp\left(\alpha_{t_{12}} V_f - \alpha_{t_{13}}(T-273.2)\right), & \text{if} \quad V_f > \alpha_{v_1} - \alpha_{v_2}(T-273.2) \\ k_{t_0}\alpha_{t_{21}} \exp\left(\alpha_{t_{22}} V_f\right), & \text{if} \quad V_f \le \alpha_{v_1} - \alpha_{v_2}(T-273.2) \end{cases}$$

$$k_p = \begin{cases} k_{p_0}, & \text{if} \quad V_f > 0.05 \\ k_{p_0}\alpha_{p_{21}} \exp\left(\alpha_{p_{22}} V_f\right), & \text{if} \quad V_f \le 0.05 \end{cases}$$

where the α's are constant parameters and M_i are the molecular weights. The total free volume $V_f = V_{f_p}\phi_p + V_{f_m}\phi_m + V$ is defined by the contributions of specific free volume of the monomer $V_m = 0.025 + \alpha_m\left(T - T_{gm}\right)$, the pol $V_{f_p} = 0.025 + \alpha_p\left(T - T_{gp}\right)$, and the solvent $V_{f_s} = 0.025 + \alpha_s\left(T - T_{gs}\right)$. The volume fractions $\phi_m = C_m M_m / \rho_m \big/ \left(\mu_1/\rho_p + C_m M_m/\rho_m + C_s M_s/\rho_s\right)$ for the monomer, $\phi_p = \mu_1/\rho_p \big/ \left(\mu_1/\rho_p + C_m M_m/\rho_m + C_s M_s/\rho_s\right)$ for the polyme $\phi_s = C_s M_s/\rho_s \big/ \left(\mu_1/\rho_p + C_m M_m/\rho_m + C_s M_s/\rho_s\right)$ for the solvent. The values of the parameters for the MMA polymeriz system are given in Table 2. Under the assumption of $F_i << F_m$, and the fact that the total volume remains constant d operation (because of the experimental design), the volumetric flowrate of the outlet stream (F) is related to the volum flowrate of the monomer stream (F_m) by $F = F_m\left(1 + \bar{\varepsilon} x_p\right)$, where x_p is the conversion factor given in terms of the concentration of polymer chains (μ_1):

$$x_p = \frac{\mu_1}{\mu_1 + M_m C_m}.$$

The volume expansion factor, $\bar{\varepsilon}$, is determined by $\bar{\varepsilon} = \phi_{m_{m_s}}\left(\overline{\rho}/\rho_p - 1\right)$, where $\phi_{m_{m_s}} = C_m M_m / \rho_m$ is the volume fracti

the monomer in the inlet stream. The other kinetic and physical parameters for the system MMA, AIBN and toluene a

Table **4**. Species balances for the monomer, initiator, solvent and dead polymer (assuming perfect mixing) and an en

balance for the reactor (assuming constant reacting mixture heat capacity c and overall heat transfer coefficient U, c_{ms}

$\rho_{ms} = \rho$, and $T\,d\rho/dt << \rho\,dT/dt$) give the set of ordinary differential equations:

$$\frac{dC_m}{dt} = R_m\left(C_m, C_i, C_s, T\right) + \left(C_{m_{m_s}} - \left(1 + \bar{\varepsilon}\, x_p\right)C_m\right)/\tau \quad (3$$

$$\frac{dC_i}{dt} = R_i\left(C_m, C_i, C_s, T\right) + \left(F_i C_{i_{i_s}} - \left(1 + \bar{\varepsilon}\, x_p\right)C_i\right)/V \quad (3$$

$$\frac{dC_s}{dt} = \left(F_m C_{s_{m_s}} + F_i C_{i_{i_s}} - F_m\left(1 + \bar{\varepsilon}\, x_p\right)C_s\right)/V \quad (3$$

$$\frac{d\mu_1}{dt} = -M_m R_m\left(C_m, C_i, C_s, T\right) - \left(\left(1 + \bar{\varepsilon}\, x_p\right)\mu_1\right)/\tau \quad (3$$

$$\frac{dT}{dt} = \alpha_0 k_p \xi_0 C_m + \alpha_1\left(T_j - T\right) + \left(T_{m_s} - \left(1 + \bar{\varepsilon}\, x_p\right)T\right)/\tau \quad (3$$

$$\frac{dT}{dt} = \alpha_2\left(T_j - T\right) + \alpha_3\left(T_\infty - T_j\right) + \alpha_4\left(P - F_{cw}\,\rho_{cw}\left(T_j - T\right)\right) \quad (3$$

where the states variables are:

- C_m: concentration of the monomer
- C_i: concentration of the initiator
- C_s: concentration of the solvent
- μ_1: is the mass concentration of the polymer chains
- T: is the reactor temperature
- T_j: is the jacket temperature.

Experimental Conditions. The aim of this example is to identify a non-linear dynamic model of the conversion fact

the polymer using the power given to the heater, the cooler flow rate and the initiator flow rate as inputs. The rest o

variables, that is, the monomer flow rate, the solvent flow rate, etc., are held constant at their steady state values. This

very nonlinear problem as can be appreciated from the Figure **7** that illustrates the steady-state conversion factor x_p v

the power P. As usual two data sets were generated to identify the system: a training and test set using a probability P_s =

The former set was used to fit the model, while the latter one was used to validate the model. The outputs were gene

integrating those inputs using the set of equations **(20)** with a sampling time of T = 1000 seconds. The DABNet mod

built to reproduce the conversion factor x_p using three inputs: the heater input power P, the cooling water inlet flow

F_{cw} and the initiator flow rate stream F_i (see *Eq*. **20**) while keeping the others constant. As a way to show the rel robustness of the approach against choosing bad the values for the Laguerre poles, those were selected t $a_1 = a_2 = a_3 = 0.5$ (just in the middle of the valid range), while the dead-times were selected to be: $d_1 = 1, d_2 = 1, d_3 = 1$ Laguerre basis terms were used for each input for the initial expansion of the linear dynamic part. For the neural networ arbitrarily choose to use 10 hidden nodes without attempting to optimize these parameters. After the application o methodology the system was reduced to a second order system for each input. Figure **8** shows the approximation result the DABNet model and for a linear model. The linear model was adjusted using a PLS (Partial Least Squares) stra being composed by a six state linear state space model. Figure **8-a** shows the DABNet approximation and the real outp the system for the training and test sets, while Figure **9-a** shows the training and test errors for the final DABNet model. comparison purposes, Figure **8-b** shows the approximation of the linear model and the real output, while Figure **9-b** sh the train and test error for the linear model. Note again that the nonlinear model performs better than the linear one. Sim considerations to the CSTR case apply to this example with respect to the steady state curve.

Note that the steady state curve of the DABNet also fits closely the steady state curve in this case. The s discrepancy in this example, however, can be found in the conversion factor range higher than 0.8. This behavior is explained by the lack of information used to train the model in this range. As can be appreciated from the Figure **7**, the used to fit the model were sparse for a conversion factor greater than 0.8. However, the DABNet model can fit this d better than any linear model. It is obvious that no matter where a straight line is located, it cannot approximate the e range.

5. Conclusions

We have shown a practical approach for identifying nonlinear models from available data. For that purpose w the DABNet (Decoupled A-B Net) structure, which is composed of a decoupled linear dynamic system followed by nonl static map. The linear dynamic system is initially spanned by a set of discrete Laguerre systems, and then cascaded w single hidden layer Perceptron. A model reduction technique (linear balancing) is performed on the hidden nodes o neural net. In that way, it is possible to identify the main time constants and to reduce the dimensionality of the Percep input. We proved that the DABNet structure is able to approximate every nonlinear, causal, discrete time invariant, mult input single-output system with fading memory. The final model consists of a linear state space system whose s decoupled by input, are mapped by a neural network. Results concerning the approximation of a pH neutralization a polymer process were presented, showing the feasibility of this approach in the approximation of typical process units.

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Appendix A

To prove the DABNet structure theorem, three Lemmas and a restatement on the Stone-Weierstrass need t presented. Moreover, recall the relation between an operator and a functional and that past (future) time is denoted by positive (non-negative) integers. Then the truncation operator $P:\boldsymbol{l}_p^\infty(\boldsymbol{Z}) \to \boldsymbol{l}_p^\infty(\boldsymbol{Z}_-)$ is defined as

$$P[\mathbf{u}](k) := \begin{cases} \mathbf{u}(k), & k \le 0 \\ \mathbf{0}, & k > 0 \end{cases}$$

while with every sequence $\mathbf{u}$ in $\boldsymbol{l}_p^\infty(\boldsymbol{Z}_-)$, we associate a causal extension sequence $\mathbf{u}_e$ in $\boldsymbol{l}_p^\infty(\boldsymbol{Z})$ defined as:

$$\mathbf{u}_e(k) = \begin{cases} \mathbf{u}(k), & k \le 0 \\ \mathbf{u}(0), & k > 0 \end{cases}$$

With each time -invariant causal operator N we associate a functional F on $\boldsymbol{l}_p^\infty(\boldsymbol{Z}_-)$ defined by $F\mathbf{u} = N\mathbf{u}$

Therefore, the operator N can be recovered from its associated functional F via

$$N\mathbf{u}(k) = FPQ^{-k}\mathbf{u}\,.$$

where Q, the delay operator, is defined as $Q^\tau[\mathbf{u}](k) := \mathbf{u}(k-\tau)$. Then, N is continuous if and only if F is, so the a equations establish a one to one correspondence between any time invariant causal continuous operator and its functio on $\boldsymbol{l}_p^\infty(\boldsymbol{Z}_-)$.

Stone-Weierstrass Theorem (*Boyd,1985).*

Suppose **E** is a compact metric space and **G** a set of continuous functionals on **E** that separates points, that is for any dis $\mathbf{u},\mathbf{v} \in \mathbf{E}$ there is a $G \in \mathbf{G}$ such that $G\mathbf{u} \neq G\mathbf{v}$. Then for any continuous functional F on **E** and given any $\varepsilon > 0$, ther functionals, $\left\{G_1^1, \cdots G_{m_1}^1, \cdots, G_1^p, \cdots G_{m_p}^p\right\} \subseteq \mathbf{G}$, $m = \sum_{i=1}^p m_i$ and a polynomial $p:\mathbf{R}^m \to \mathbf{R}$, such that for all $\mathbf{u} \in \mathbf{E}$

$$\left|F\mathbf{u} - p\left(G_1^1\mathbf{u}, \cdots, G_{m_1}^1\mathbf{u}, \cdots, G_1^p\mathbf{u}, \cdots, G_{m_p}^p\mathbf{u}\right)\right| < \varepsilon$$

The reason for the group indexing, which is not necessary for a general statement of the Stone-Weierstrass theorem, become apparent in Lemma 2 when we consider each block with a Laguerre operator. In addition, the fading me definition is stated and three lemmas are necessary before the theorem can be proved.

Fading Memory: *A*n operator $N:\boldsymbol{l}_p^\infty(\boldsymbol{Z}) \to \boldsymbol{l}_p^\infty(\boldsymbol{Z})$ has fading memory on a subset $\boldsymbol{K}_-$ of $\boldsymbol{l}_p^\infty(\boldsymbol{Z}_-)$ if there is a decre sequence $w(k): \boldsymbol{Z}_+ \to (0,1]$, $\lim_{k\to\infty} w(k) = 0$, such that for each $\mathbf{u} \in \boldsymbol{K}_-$ and $\varepsilon > 0$ there is $\delta > 0$ such that for all $\mathbf{v} \in \boldsymbol{K}_-$

$$\sup_{k\le 0}\left\{\left|\mathbf{u}(k) - \mathbf{v}(k)\right| w(-k)\right\} < \delta \to \left|N\mathbf{u}(0) - N\mathbf{v}(0)\right| < \varepsilon\,.$$

Lemma 1: *The sub-space* $K_- \equiv \left\{ \mathbf{u} \in l_p^\infty(Z_-) \middle| 0 < \|\mathbf{u}\| \le c_1 \right\}$, *is compact with the* $\|\cdot\|_w$ *norm defined for any decreasing sequ*

w: $\boldsymbol{Z}_+ \to (0,1]$, $\lim_{k\to\infty} w(k) = 0$:

$$\|\mathbf{u}\|_w := \sup_{k\le 0}\{|\mathbf{u}(k)|w(-k)\}. \qquad (A$$

Proof: Let $\mathbf{u}^{(n)}$ be any sequence in K_-. We will find $\mathbf{u}^{(0)} \in K_-$ and a subsequence of $\mathbf{u}^{(n)}$ converging in the $\|\cdot\|_w$ nor

$\mathbf{u}^{(0)}$. It is well known that K_- is not compact in $l_p^\infty(\mathbf{Z}_-)$ with the usual supremum norm $\|\cdot\|_\infty$ (Kolmogorov, 1980). For

l, let be $K_-[-l,0]$ the restriction of K_- to $[-l,0]$. $K_-[-l,0]$ is uniformly bounded by c_1 and is composed of a finite s

values, hence compact in $l_p^\infty[-l,0]$. Since $K_-[-l,0]$ is compact for every *l*, we can find a subsequence $\mathbf{u}^{(n_m)}$ of $\mathbf{u}^{(n)}$ a

$\mathbf{u}^{(0)} \in K_-[-l,0]$ along which $\mathbf{u}^{(n_m)}$ converges:

$$\sup_{-l\le k\le 0}\left|\mathbf{u}^{(n_m)}(k) - \mathbf{u}^{(0)}(k)\right| \to 0 \text{ as } m\to\infty. \qquad (A$$

Now, for some $\varepsilon > 0$, since $w(k) \to 0$ as $k \to \infty$, we can find a $k_0 > 0$ such that $w(k_0) \le \varepsilon/c_1$. Since $\mathbf{u}^{(n_m)}$, $\mathbf{u}^{(0)} \in K_-$

have that

$$\sup_{k\le -k_0}\left\{\left|\mathbf{u}^{(n_m)}(k) - \mathbf{u}^{(0)}(k)\right| w(-k)\right\} \le 2c_1 w(k_0) < \varepsilon \qquad (A$$

Now from Equation Ap3 we can find m_1 such that

$$\sup_{-k_0<k\le 0}\left|\mathbf{u}^{(n_m)}(k) - \mathbf{u}^{(0)}(k)\right| < \varepsilon \text{ for } m > m_1 \qquad (A$$

so by Equation Ap3 and Equation Ap5 we can conclude that $\left\|\mathbf{u}^{(n_m)} - \mathbf{u}^{(0)}\right\|_w < \varepsilon$, $m > m_1$ which proves that K_- is compa

Lemma 2. *The set of functionals* $\{G_i^j\}$ *associated to the discrete Laguerre Operators are continuous with respect t*

norm $\|\cdot\|_w$, *that is, given any* $\varepsilon > 0$ *there exists* $\delta > 0$ *such that* $\|\mathbf{u} - \mathbf{v}\|_w < \delta \Rightarrow \left|G_i^j\mathbf{u} - G_i^j\mathbf{v}\right| < \varepsilon$.

Proof. Every Laguerre system is associated with an operator that defines the input-output as: $L_i^j\mathbf{u}(k) = L_i^j u_j(k)$. With

of the Laguerre operators $L_i^j(\cdot)$ we associate a functional defined by $G_i^j u_i = \sum_{k=0}^\infty l_j^i(k)u_i(-k)$, such that the applicatio

the functional to a sequence is performed as $G_i^j\mathbf{u} = G_i^j u_j$. Consider the functional $G_i^j(\cdot)$ associated with $L_i^j(\cdot)$, so g

$\varepsilon > 0$, chose $\delta > 0$ such that:

$$\left|u_j - v_j\right| < \delta \qquad \Rightarrow \qquad \left|G_i^j u_j - G_i^j v_j\right| < \varepsilon \qquad (A$$

This is possible due to the continuity of the one-dimensional Laguerre operators with respect to the weighted norm as sh
in (Sentoni et al, 1996). Therefore, from Equation Ap6 and the definition of the functionals

$$\|\mathbf{u}-\mathbf{v}\|_w < \delta \Rightarrow \left|u_j - v_j\right|_w < \delta \Rightarrow \left|G_i^j\mathbf{u} - G_i^j\mathbf{v}\right| = \left|G_i^j u_j - G_i^j v_j\right| < \varepsilon \qquad (A$$

which proves Lemma 2■

Lemma 3. *The functionals* $\left\{G_i^j\right\}$ *separate points in* $l_p^\infty(\mathbf{Z}_-)$, *that is, for any distinct* $\mathbf{u},\mathbf{v} \in l_p^\infty(\mathbf{Z}_-)$ *there is a* $G_i^j \in \mathbf{G}$
that $G_i^j\mathbf{u} \neq G_i^j\mathbf{v}$.

Proof. Suppose $\mathbf{u},\mathbf{v} \in l_p^\infty(\mathbf{Z}_-)$ are equal except for the *j*-th component. Then by the functionals' definition

$$G_i^j\mathbf{u} \neq G_i^j\mathbf{v} \Leftrightarrow G_i^j u_j \neq G_j^i v_j \qquad (A$$

It known from one dimensional theory (Sentoni et al, 1996) that for any distinct $u_j, v_j \in l^\infty(\mathbf{Z}_-)$ there is a G_i^j such
$G_i^j u_j \neq G_i^j v_j$; this result together with Equation Ap8 proves Lemma 3■

Approximation Theorem

Now given $\varepsilon > 0$, Lemmas 1, 2, 3 together with the Stone-Weierstrass theorem imply that given any continuous functio
on K_-, there is a polynomial $p:\mathbf{R}^m \to \mathbf{R}$, such that for all $\mathbf{u} \in K_-$

$$\left|F\mathbf{u} - p\left(G_1^1\mathbf{u},\cdots,G_{m_1}^1\mathbf{u},\cdots,G_1^p\mathbf{u},\cdots,G_{m_p}^p\mathbf{u}\right)\right| < \varepsilon \qquad (A$$

Because the Laguerre systems are continuous and act on a bounded space, the $G_i^j\mathbf{u}$ are bounded real intervals. So
polynomial *p* can be replaced by any static model that acts as a universal approximator on a bounded input space
example, a Neural Net. In other words Equation Ap9 can be replaced by

$$\left|F\mathbf{u} - NN\left(G_1^1\mathbf{u},\cdots,G_{m_1}^1\mathbf{u},\cdots,G_1^p\mathbf{u},\cdots,G_{m_p}^p\mathbf{u}\right)\right| < \varepsilon$$

Using Equation Ap1 a time invariant causal operator *N* can be recovered from its associated functional, so let $\mathbf{u} \in K$
$k \in \mathbf{Z}$, so $PQ^{-k}\mathbf{u} \in K_-$, hence

$$\left|FPQ^{-k}\mathbf{u} - NN\left(G_1^1PQ^{-k}\mathbf{u},\cdots,G_{m_1}^1PQ^{-k}\mathbf{u},\cdots,G_1^pPQ^{-k}\mathbf{u},\cdots,G_{m_p}^pPQ^{-k}\mathbf{u}\right)\right| < \varepsilon .$$

Since the last equation is true for all $k \in \mathbf{Z}$, we conclude that for all $\mathbf{u} \in K_-$, $\left\|N\mathbf{u} - \widehat{N}\mathbf{u}\right\| < \varepsilon$. This completes the proof♥

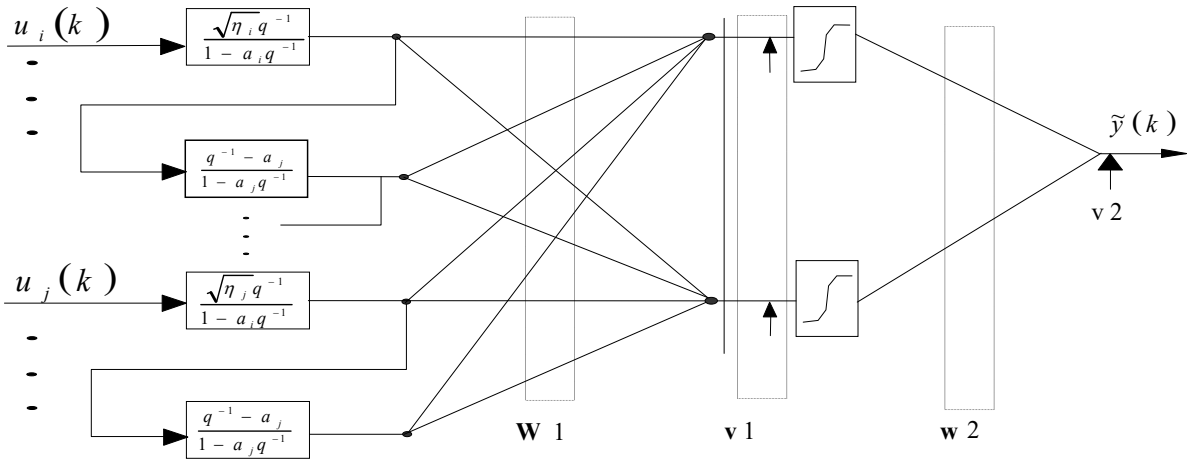
Table 1. pH *process Nominal conditions*		
tanks volume	V	1000 l
tank level	h_1, h_2	10 cm
valve coefficient 1	c_{V1}	188.5 $cm^{-1/2}$ / min
valve coefficient 2	c_{V2}	351.3 $cm^{-1/2}$ / min
acid flowrate 1	Q_1	41 l/min
acid flowrate 2	Q_2	40 l/min
base flowrate 3	Q_3	515 l/min
acid flowrate 5	Q_5	50 l/min
acid flowrate 6	Q_6	31 l/min
base flowrate 7	Q_7	515 l/min
HAC concentration	C_A	0.32 mol/l
N_aOH concentration	C_B	0.05 mol/l
HAC dissoc. Csnt	K_A	1.8×10^{-5}
N_aOH dissoc. Csnt	K_B	1.0
H_2O dissoc. Csnt	K_w	1.0×10^{-14}
effluent pH value	pH_i	i:1,2 =7.0
sampling interval	t_s	0.1 min

Table 2: *Parameters of the system*		
T_{cw}	$= 2.9520 \times 10^{2}$	K
T_{∞}	$= 2.9520 \times 10^{2}$	K
T_{ms}	$= 2.9520 \times 10^{2}$	K
c_{w}	$= 4.2000 \times 10^{0}$	$kJ\ kg^{-1}\ K^{-1}$
ρ_{w}	$= 1.0000 \times 10^{3}$	$kg\ m^{3}$
P_{max}	$= 3.0000 \times 10^{0}$	$kJ\ s^{-1}$
F_{cwmax}	$= 2.5500 \times 10^{-4}$	$m^{3}s^{-1}$
F_{imax}	$= 2.3000 \times 10^{-8}$	$m^{3}s^{-1}$
F_{m}	$= 2.8000 \times 10^{-7}$	$m^{3}s^{-1}$
V	$= 1.2000 \times 10^{-3}$	m^{3}
α_{1}	$= 3.8000 \times 10^{-3}$	s^{-1}
α_{2}	$= 8.0000 \times 10^{-4}$	s^{-1}
α_{3}	$= 3.7000 \times 10^{-4}$	s^{-1}
α_{4}	$= 6.6400 \times 10^{-2}$	$K\ kJ^{-1}$

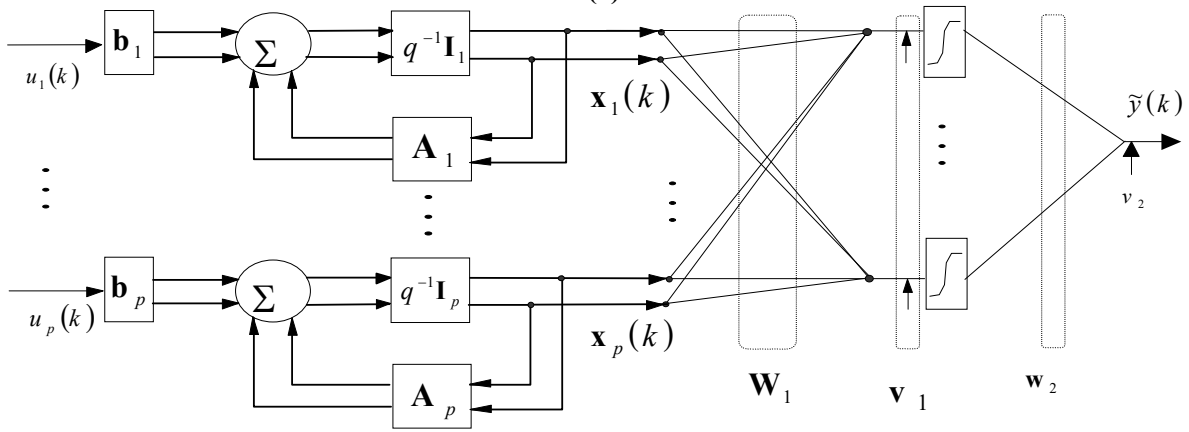
Table 3. *Gel and glass effect model parameters*	
$\alpha_{t_{11}} = 1.0575 \times 10^{-1}$	$\alpha_{t_{12}} = 1.7150 \times 10^{1}$
$\alpha_{t_{13}} = 1.7150 \times 10^{-2}$	$\alpha_{t_{21}} = 2.3000 \times 10^{-6}$
$\alpha_{t_{22}} = 7.5000 \times 10^{1}$	$\alpha_{p_{21}} = 7.1000 \times 10^{-5}$
$\alpha_{p_{22}} = 1.7153 \times 10^{-2}$	$\alpha_{v_1} = 1.8560 \times 10^{-1}$
$\alpha_{v_{22}} = 2.9650 \times 10^{-4}$	$\alpha_{p} = 4.8000 \times 10^{-4}$
$\alpha_{s} = 1.0000 \times 10^{-3}$	$\alpha_{m} = 1.0000 \times 10^{-3}$

Table 4: *Operating conditions*	
$C_{i_{is}} = 2.6340 \times 10^{-1}$	$kgmole\ m^{-3}$
$C_{s_{is}} = 9.9860 \times 10^{0}$	$kgmole\ m^{-3}$
$C_{s_{ms}} = 5.8810 \times 10^{0}$	$kgmole\ m^{-3}$
$C_{m_{ms}} = 3.9840 \times 10^{0}$	$kgmole\ m^{-3}$

Table 5: *Physical and kinetics parameters*		
k_l	$= Z_l \exp\left(\frac{-E_l}{RT}\right)$	$l = f_m, i, t_0, P_0$
Z_{t_0}	$= 9.8000 \times 10^{7}$	$m^3 kgmole^{-1} s^{-1}$
E_{t_0}	$= 2.9442 \times 10^{3}$	$kJ\ kgmole^{-1}$
Z_{P_0}	$= 4.9167 \times 10^{5}$	$m^3 kgmole^{-1} s^{-1}$
E_{P_0}	$= 1.8283 \times 10^{4}$	$kJ\ kgmole^{-1}$
Z_{f_m}	$= 4.6610 \times 10^{9}$	$m^3 kgmole^{-1} s^{-1}$
E_{f_m}	$= 7.4479 \times 10^{4}$	$kJ\ kgmole^{-1}$
Z_i	$= 1.0533 \times 10^{15}$	s^{-1}
E_i	$= 1.2877 \times 10^{5}$	$kJ\ kgmole^{-1}$
T_{gp}	$= 3.8720 \times 10^{2}$	K
T_{gm}	$= 1.6720 \times 10^{2}$	K
T_{gs}	$= 1.1320 \times 10^{2}$	K
c	$= 1.8150 \times 10^{0}$	$kJ\ kg^{-1}\ K^{-1}$
f	$= 5.8000 \times 10^{-1}$	
m	$= 8.6460 \times 10^{0}$	kg
R	$= 8.3450 \times 10^{0}$	$kJ\ kgmole^{-1} K^{-1}$
$\overline{\rho}_m$	$= 8.8570 \times 10^{2}$	$kg\ m^{-3}$
$\overline{\rho}_s$	$= 8.1550 \times 10^{2}$	$kg\ m^{-3}$
ρ_p	$= 1.1660 \times 10^{3}$	$kg\ m^{-3}$
M_m	$= 1.0012 \times 10^{2}$	$kJ\ kgmole^{-1}$
M_s	$= 9.2140 \times 10^{2}$	$kJ\ kgmole^{-1}$
M_i	$= 1.6421 \times 10^{2}$	$kJ\ kgmole^{-1}$
ΔH_P	$= 5.7800 \times 10^{4}$	$kJ\ kgmole^{-1}$


$u_i(k)$
$\frac{\sqrt{\eta_i}\,q^{-1}}{1-a_i q^{-1}}$
$\frac{q^{-1}-a_j}{1-a_j q^{-1}}$
$u_j(k)$
$\frac{\sqrt{\eta_j}\,q^{-1}}{1-a_i q^{-1}}$
$\frac{q^{-1}-a_j}{1-a_j q^{-1}}$
$\tilde{y}(k)$
v 2
W 1
v 1
w 2

(a)


$\mathbf{b}_1$
$u_1(k)$
Σ
$q^{-1}\mathbf{I}_1$
$\mathbf{A}_1$
$\mathbf{x}_1(k)$
$\mathbf{b}_p$
$u_p(k)$
Σ
$q^{-1}\mathbf{I}_p$
$\mathbf{A}_p$
$\mathbf{x}_p(k)$
$\tilde{y}(k)$
v_2
$\mathbf{W}_1$
$\mathbf{v}_1$
$\mathbf{w}_2$

(b)

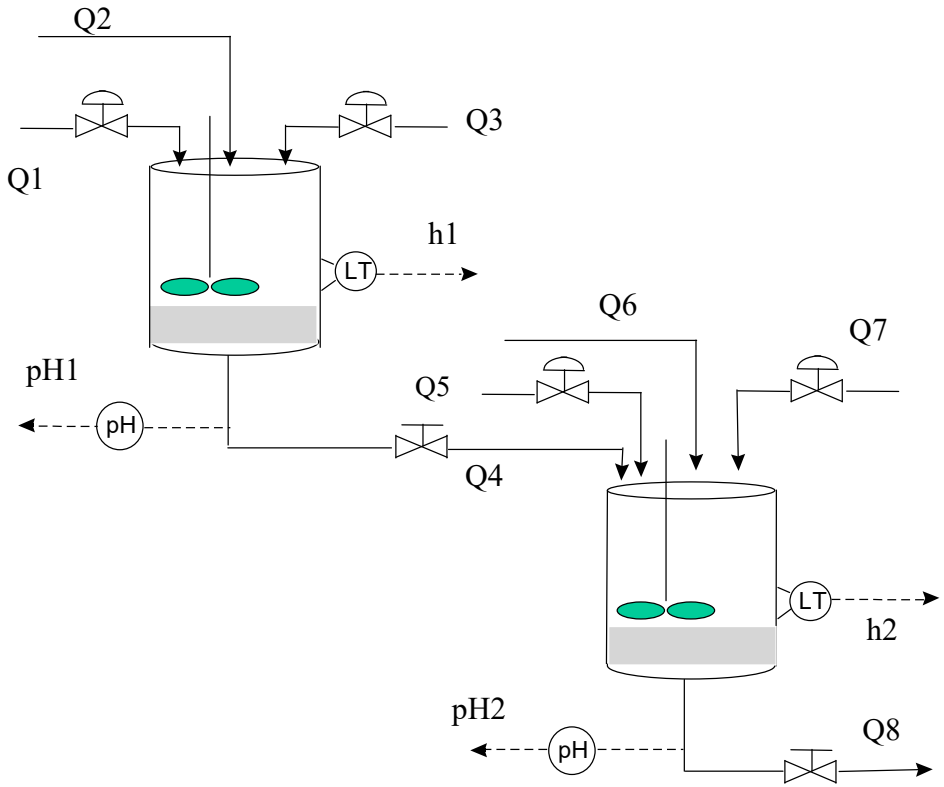


Figure 2

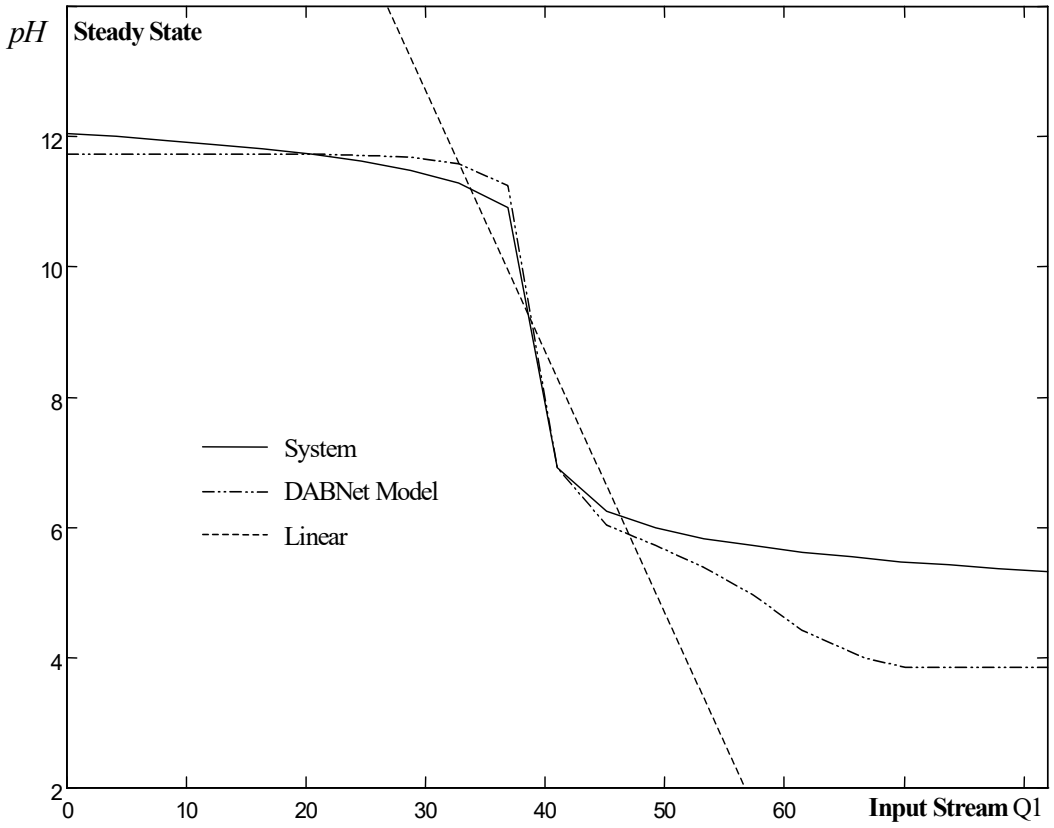
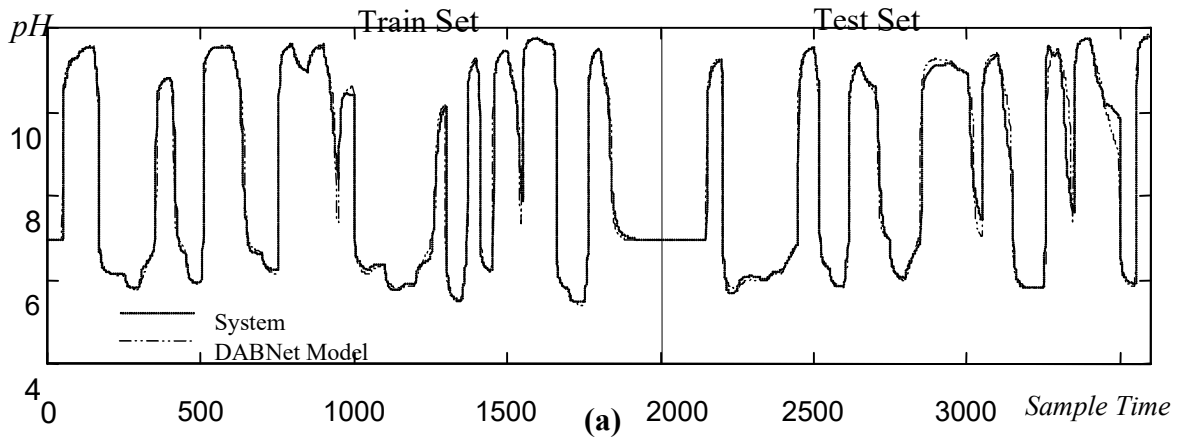
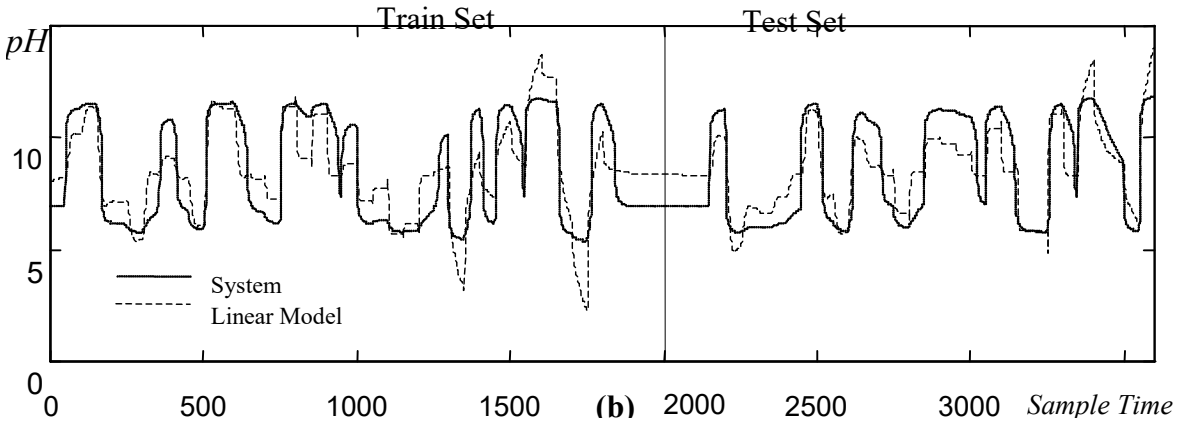

pH
Steady State
12
10
8
6
4
2
0
10
20
30
40
50
60
Input Stream Q1
System
DABNet Model
Linear

Figure 3


pH
Train Set
Test Set
10
8
6
4
System
DABNet Model
0
500
1000
1500
2000
2500
3000
Sample Time

(a)


pH
Train Set
Test Set
10
5
0
System
Linear Model
0
500
1000
1500
2000
2500
3000
Sample Time

(b)

Figure 4

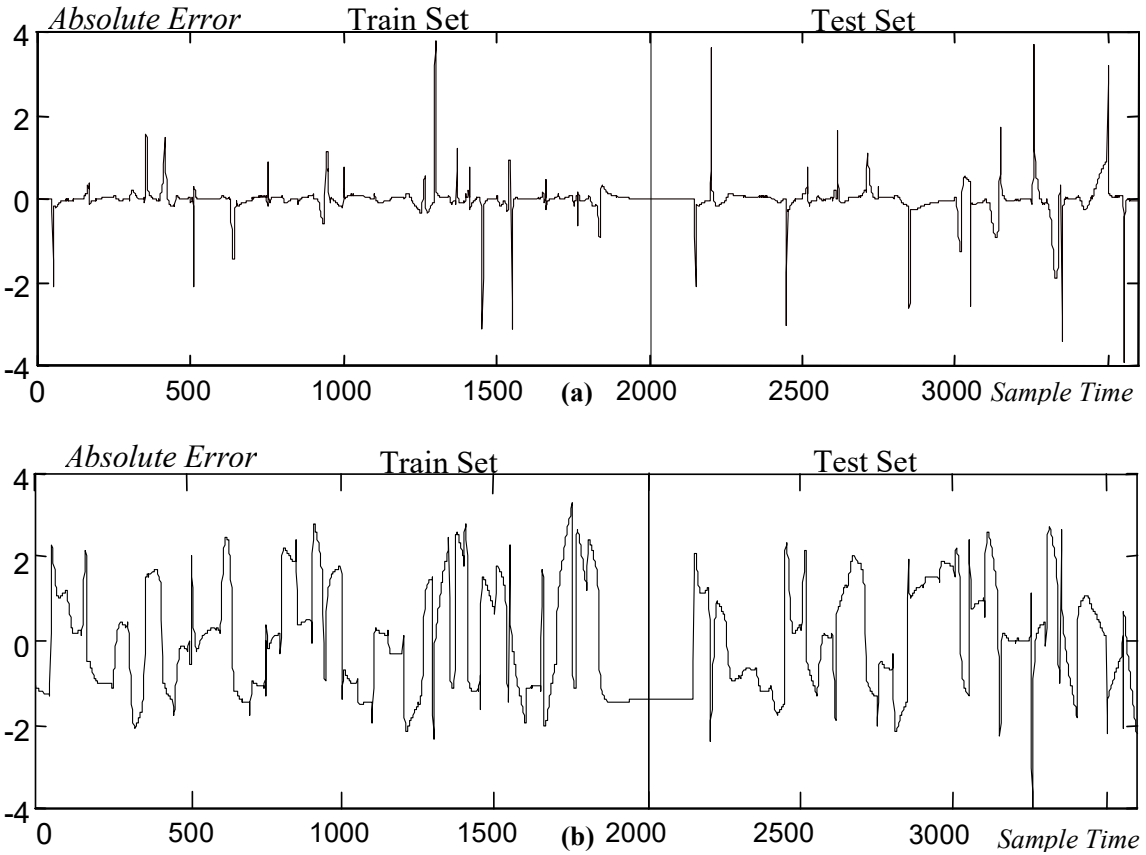

Absolute Error
Train Set
Test Set
4
2
0
-2
-4
0
500
1000
1500
(a)
2000
2500
3000
Sample Time
Absolute Error
Train Set
Test Set
4
2
0
-2
-4
0
500
1000
1500
(b)
2000
2500
3000
Sample Time

Figure 5

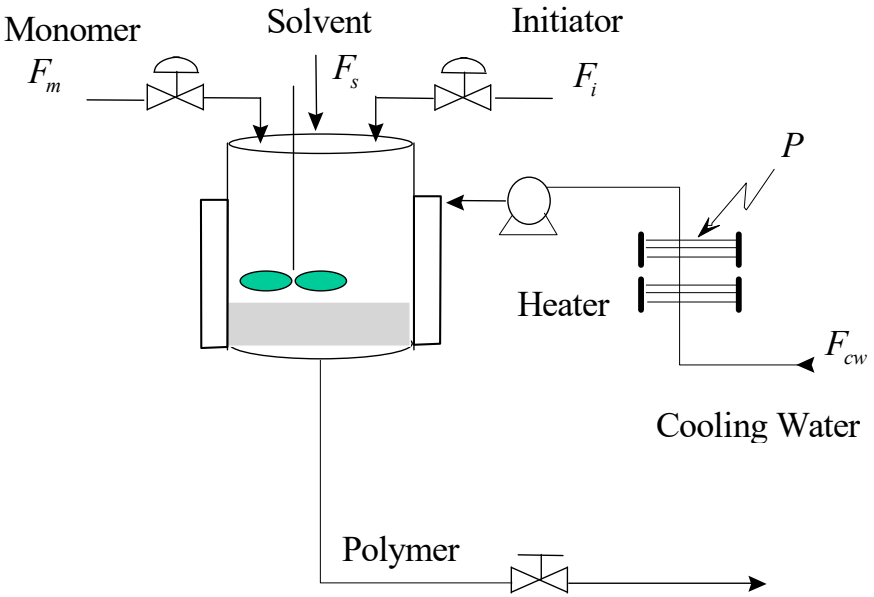

Monomer
F_m
Solvent
F_s
Initiator
F_i
P
Heater
F_{cw}
Cooling Water
Polymer

Figure 6

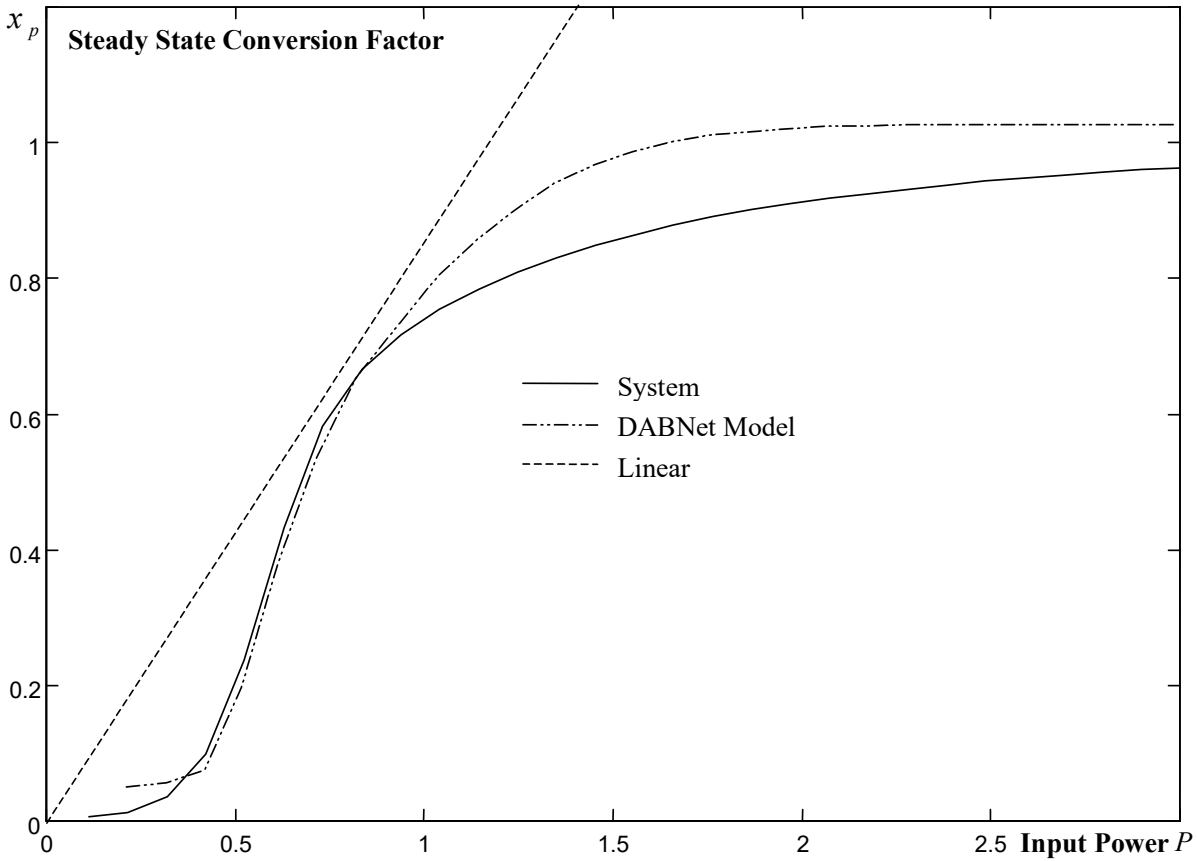

x_p
Steady State Conversion Factor
1
0.8
0.6
0.4
0.2
0
0
0.5
1
1.5
2
2.5
Input Power P
System
DABNet Model
Linear

Figure 7

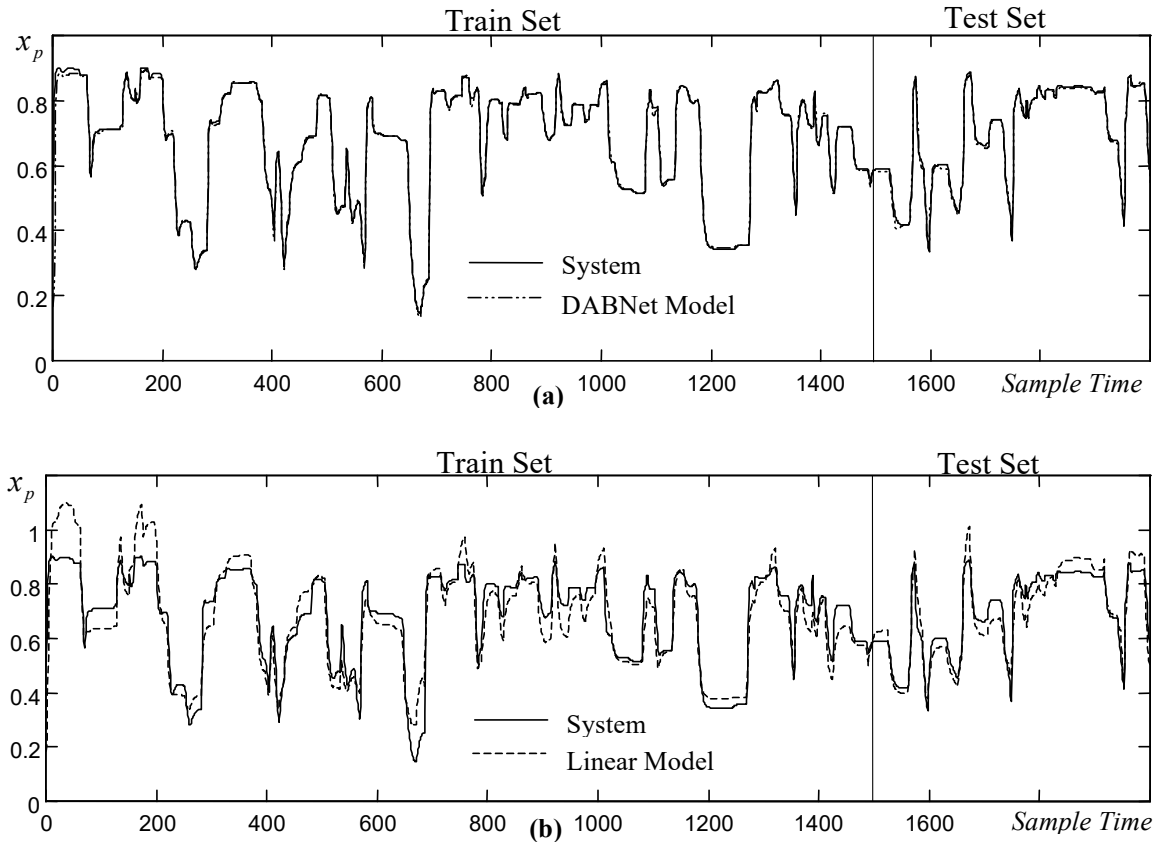
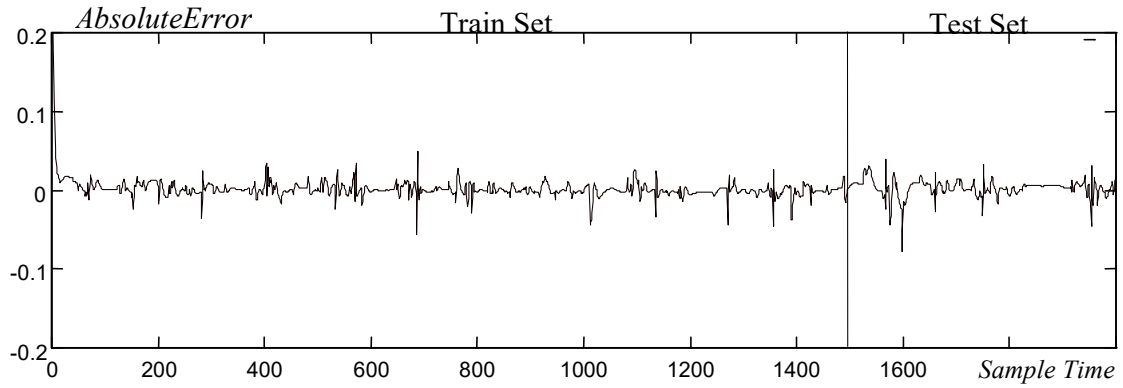
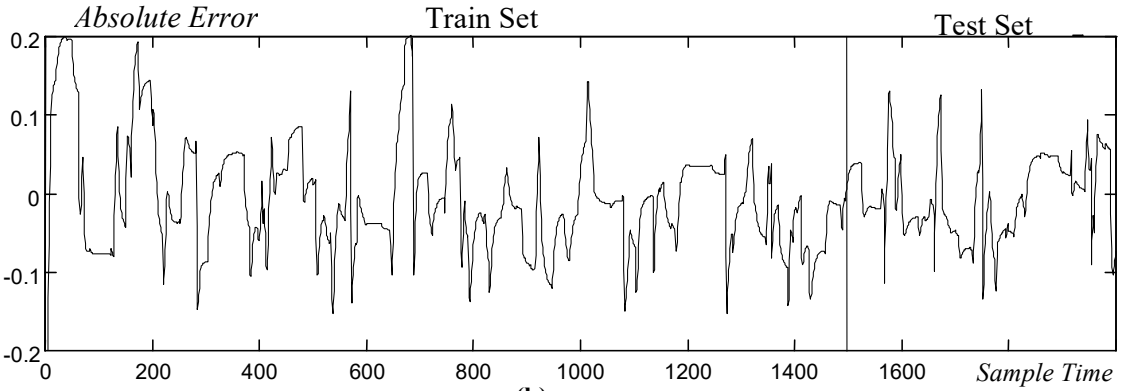

Train Set
Test Set
x_p
0.8
0.6
0.4
0.2
0
0
200
400
600
800
1000
1200
1400
1600
Sample Time
System
DABNet Model
(a)
Train Set
Test Set
x_p
1
0.8
0.6
0.4
0.2
0
0
200
400
600
800
1000
1200
1400
1600
Sample Time
System
Linear Model
(b)

Figure 8


AbsoluteError
Train Set
Test Set
0.2
0.1
0
-0.1
-0.2
0
200
400
600
800
1000
1200
1400
1600
Sample Time

(a)


Absolute Error
Train Set
Test Set
0.2
0.1
0
-0.1
-0.2
0
200
400
600
800
1000
1200
1400
1600
Sample Time

(b)

Figure 9

Figures Captions

Figure 1: Proposed *DABNet* structure

a) Laguerre' Systems Initialization.

b) Final Reduced Structure.

Figure 2: Schematic Diagram of the pH neutralization system.

Figure 3: Steady-State Behavior of the pH neutralization system the DABNet Model and the Linear Model.

Figure 4: Approximation of the ***pH*** discriminated by Train and Test set for the CSTR Example.

a) Approximation of the ***pH*** by the *DABNet* model.

b) Approximation of the ***pH*** by the *Linear* model.

Figure 5: Absolute Error of the ***pH*** Approximation discriminated by Train and Test set in the CSTR example.

a) *DABNet* model. The norm of the error is: 12.383 for the *train set* and 18.580 for the *test set*.

b) *Linear* model. The norm of the error is: 60.223 for the *train set* and 50.186 for *the test set*.

Figure 6: Schematic Diagram of the Polymerization System.

Figure 7: Steady-State Behavior of the Polymerization System, the DABNet Model and the Linear Model.

Figure 8: Conversion Factor Approximations discriminated by Train and Test set for the Polymer example.

a) Approximation of $\boldsymbol{x_p}$ by the *DABNet* model.

b) Approximation of $\boldsymbol{x_p}$ by the *Linear* model.

Figure 9: Absolute Error of the $\boldsymbol{x_p}$ Approximation discriminated by Train and Test set in the Polymer Example.

a) *DABNet* model. The norm of the error is: 1.263 for the *train set* and 0.251 for the *test set*

b) *Linear* model. The norm of the error is: 2.862 for the *train set* and 1.122 for the *test set*.